

Data Collection Guide — Valuing a No-Earnings Battery Company

Built as a public-data portfolio project for GitHub

A practical guide to the data you need to value a development-stage battery company — using entirely free, public Australian sources, so the whole thing is reproducible and portfolio-ready.

Educational resource, not investment advice. I'm not a licensed financial adviser. Verify every figure and source date before relying on it.

The big idea: why this works as a portfolio piece

A real evaluation of a *specific* company needs private data (their actual pipeline, costs, contracts) from a data room. **You can't put that on GitHub.** But here's the elegant part:

The entire valuation *methodology* can be demonstrated using only free public data and a realistic illustrative (synthetic) company. The portfolio shows you can *build the model and source the inputs* — which is exactly what a recruiter in credit risk, infrastructure, or PE wants to see.

So the project structure is:

1. **Build an illustrative battery developer** — a made-up but realistic company with a pipeline of small NSW/VIC projects. You construct this *from public data* (real project sizes, real locations, real cost benchmarks), so it's grounded, not fantasy.
2. **Source every model input from public, free sources** (this guide shows where).
3. **Run the five valuation methods** (risk-adjusted NAV, \$/MW, scenarios, VC method, comps).
4. **Document every source and assumption** so it's fully reproducible.

A recruiter clicking your repo sees: real data sourcing, a defensible risk-adjusted valuation, clean Python/Excel, and clear documentation. That's a strong signal.

A portfolio hook tailored to you: the "probability of success at each development gate" (will it get approved? connected? sold?) is structurally **the same idea as a PD — probability of default — model**, your exact specialty. You can build those success probabilities with the same statistical rigour you apply to credit PD work, sourced from public approval and connection data. **That bridge — credit-risk modelling applied to infrastructure development — is a standout story for your job search.** Make it the centrepiece of the README.

What the valuation needs, and the public source for each input

Here's every input the five methods require, mapped to a free public source. The clever moves are in the "normally private → public proxy" rows.

Inputs for Method 1 — Risk-adjusted NAV (the core model)

Input	Normally from...	Free public source you'll use instead	How
The project pipeline (list of projects, MW/MWh, location, stage)	Company data room	AEMO Generation Information file	Download the spreadsheet (free, ~quarterly). It lists every existing/committed/proposed project in the NEM. Filter for NSW/VIC batteries to build a realistic illustrative pipeline.
Sale price per project (\$/MW when de-risked)	Negotiation / comps	Energy trade press deal announcements	Energy-Storage.news, pv magazine Australia, RenewEconomy, ESD News report battery sales with sizes — derive \$/MW. (See Method 5.)
Development + build cost (\$/kW or \$/kWh)	Company estimates	CSIRO GenCost report (free, authoritative)	The annual GenCost report (CSIRO + AEMO) gives Australian battery capital costs. This is the gold-standard free cost source.
Cost assumptions cross-check	Company	AEMO ISP input/assumptions workbook	The ISP publishes its cost and technology assumptions — free, citable.
Probability of success at each gate	Company track record	NSW Planning Portal + AEMO Connections Scorecard (see below)	Compute these yourself from public approval and connection data — the heart of the portfolio.
Timing (years to each milestone)	Company plan	NSW Planning Portal (approval durations) + AEMO Connections Scorecard (connection durations)	Both publish actual timelines — measure lodgement-to-determination yourself.
Discount rate	Investor's cost of capital	RBA (risk-free rate) + public renewable valuation guidance	RBA publishes the cash rate / bond yields. Add a development-stage risk premium (industry guidance: 20%+ for early-stage development phases).

Input	Normally from...	Free public source you'll use instead	How
Company overhead	Company accounts	Illustrative assumption	State it clearly as an assumption.

The clever part: sourcing "probability of success" from public data

This is normally a company's private track record. You build it from public data — and it's the most impressive, most *you* part of the project.

Gate	The question	Public data to compute it	How
Planning approval	What % of BESS projects get approved, and how long does it take?	NSW Planning Portal (Major Projects / Application Tracker)	Count approved vs refused/withdrawn BESS projects → an approval rate. Measure lodgement-to-determination dates → a timeline distribution.
Grid connection	What % connect successfully, and how long?	AEMO Connections Scorecard + Key Connection Information	AEMO publishes how projects move through the connection process, approval rates, and typical durations.
Reaching financial close / sale	What share of the pipeline actually completes?	AEMO ISP / Generation Information (committed vs proposed ratios) + trade press	Compare how many "proposed" projects become "committed/operational" → an attrition rate.

The PD-model parallel to highlight in your portfolio: treat each gate as a stage in a **survival model** — at each gate a project either advances or "defaults" (stalls). Cumulative probability of success = product of the stage probabilities. This is conceptually identical to a multi-period PD/survival curve. Build it with public data and the same statistical care you'd give a credit model. That's the headline skill the project demonstrates.

Inputs for Method 2 — \$/MW benchmark

Input	Free public source	How
\$/MW comps by development stage	Trade press deal announcements + renewable valuation blogs (Green Giraffe, Phoenix Strategy)	Collect recent battery deal \$/MW figures; note the stage (early / ready-to-build / built)
Total pipeline MW	AEMO Generation Information (your illustrative pipeline)	Sum the MW

Inputs for Method 3 — Scenario weighting

Input	Free public source	How
Upside / base / downside values	Outputs of Method 1 under different assumptions	Re-run your NAV model with optimistic/pessimistic inputs
Scenario probabilities	Your judgment, informed by the public approval/connection data	Document the reasoning

Inputs for Method 4 — VC method

Input	Free public source	How
Exit value	Pipeline MW × \$/MW at exit (from comps)	Method 5 feeds this
Target return (hurdle)	Public renewable development return benchmarks	Industry guidance: development-stage returns of ~20%+; state your chosen hurdle
Risk-free rate	RBA	Current government bond yield

Inputs for Method 5 — Comparable transactions (feeds everything)

Input	Free public source	How
Recent battery project/platform sales	Energy-Storage.news , pvmagazine Australia , RenewEconomy , ESD News	Search the last 12 months of battery M&A announcements; record size, stage, contract status, price
Government green-capital deals	CEFC + Energy Security Corporation websites	Their investment announcements signal buyer appetite and sometimes pricing

The free public sources, in one place

Everything above, consolidated. All free.

Source	What it gives you for the valuation
AEMO Generation Information file	The project pipeline (build your illustrative company from it)
AEMO Connections Scorecard / KCI	Grid-connection success rates and timelines → probability + timing inputs
AEMO Integrated System Plan (ISP)	Cost assumptions, storage need, attrition (proposed→committed)
AEMO Quarterly Energy Dynamics (QED)	Current market context
CSIRO GenCost report	Authoritative Australian battery capital costs
NSW Planning Portal (Major Projects / Application Tracker)	Approval rates, approval timelines, project details → probability + timing inputs
Victorian planning system (Engage Victoria / DTP)	VIC approval data
RBA	Risk-free rate for the discount rate
CEFC + Energy Security Corporation	Buyer-demand signals, some deal info
Energy trade press (Energy-Storage.news, pv magazine Australia, RenewEconomy, ESD News)	\$/MW comps, exit values, recent deals

One honesty note on comps: the richest deal databases (BloombergNEF, Enerdatix, Mergermarket) are *paid*. But enough deals are announced in the *free* trade press to build a credible comps table. Note in your README that comps rely on publicly reported deals — that's a legitimate, transparent limitation, not a weakness.

How to build it as a portfolio project (step by step)

A checklist to take it from data to a repo a recruiter admires:

☐ **Step 1 — Build the illustrative pipeline.** Pull 5-10 realistic small NSW/VIC battery projects from the AEMO Generation Information file (real sizes/locations), or design a

- synthetic pipeline grounded in those real figures. Document that it's illustrative.
- ❑ **Step 2 — Source the cost inputs.** Get battery \$/kWh capital cost from CSIRO GenCost. Cite the exact report and year.
 - ❑ **Step 3 — Build the success-probability ("PD-style") model.** From the NSW Planning Portal and AEMO Connections Scorecard, compute approval and connection success rates and timelines. This is your statistical centrepiece — show the working.
 - ❑ **Step 4 — Build the \$/MW comps table.** From free trade press, collect recent battery deal prices by stage. Cite each source and date.
 - ❑ **Step 5 — Set the discount rate.** RBA risk-free rate + a documented development risk premium.
 - ❑ **Step 6 — Build the risk-adjusted NAV model** (Method 1) in Python or Excel: per project → margin → × probability → discounted → summed.
 - ❑ **Step 7 — Cross-check** with the \$/MW benchmark (Method 2), scenario weighting (Method 3), and VC method (Method 4).
 - ❑ **Step 8 — Produce a valuation *range*** and a short written conclusion.
 - ❑ **Step 9 — Document everything:** a clear README, a source log (every input + source + date), and reproducible code.
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What makes this repo impressive (recruiter's eye)

Frame these deliberately — they map to what credit-risk, infrastructure, and PE employers screen for:

What you show	Why it lands
Public-data sourcing	Proves you can find and use authoritative data independently
A PD-style probability model	Bridges your credit-risk expertise into a new domain — your unique angle
Risk-adjusted NAV (rNPV)	A genuinely institutional valuation technique (used in biotech, infra, PE)
Triangulation + a range	Shows mature judgment, not false precision
Full reproducibility + source log	Signals the governance and documentation discipline you already practise (MRC-ready thinking)
An honest limitations section	Demonstrates the conservative, defensible framing you value

README opening worth writing: something like *"This project values an illustrative development-stage battery storage company using only free, public Australian data. It*

applies a risk-adjusted NAV (rNPV) framework, with project success probabilities modelled using a survival/PD-style approach sourced from public planning-approval and grid-connection data." That single sentence tells a recruiter exactly what skills you're demonstrating.

A simple suggested repo structure

```
battery-developer-valuation/  
├── README.md           ← the story + methodology + sources  
├── data/  
│   ├── pipeline.csv    ← illustrative projects (from AEMO GenInfo)  
│   ├── costs.csv       ← from CSIRO GenCost  
│   ├── comps.csv       ← $/MW deals from trade press  
│   └── probabilities.csv ← computed approval/connection rates  
├── notebooks/  
│   ├── 01_probability_model.ipynb ← the PD-style success model  
│   ├── 02_rnpv_valuation.ipynb    ← the core NAV model  
│   └── 03_triangulation.ipynb     ← $/MW, scenarios, VC method  
├── src/                   ← reusable functions  
├── outputs/  
│   └── valuation_summary.md ← the range + conclusion  
└── SOURCES.md             ← every input, its source, and date
```

Check yourself

1. Why can you build this on public data when a real evaluation needs private data? *(Answer: the portfolio demonstrates the methodology on an illustrative company; the inputs have public proxies.)*
 2. Where do the "probability of success" numbers come from? *(Answer: computed from public NSW Planning Portal approval data and AEMO connection data.)*
 3. What's the free, authoritative source for battery costs? *(Answer: the CSIRO GenCost report.)*
 4. Why is the probability model your strongest portfolio angle? *(Answer: it applies credit-risk PD/survival modelling — your specialty — to infrastructure, bridging your background into PE/infra.)*
 5. What honest limitation should the README state? *(Answer: comps rely on publicly reported deals, since the richest deal databases are paid.)*
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All sources named (AEMO Generation Information, Connections Scorecard, ISP, QED; NSW Planning Portal; CSIRO GenCost; RBA; CEFC; energy trade press) are free and verified

current as of mid-2026; confirm titles and dates at each source. Methods (risk-adjusted NPV, \$/MW multiples, scenario weighting, VC method, comps) reflect standard valuation theory and current renewable-development practice. This is an educational resource for building your own analysis, not investment advice; I am not a licensed financial adviser. Any illustrative company and figures must be clearly labelled as illustrative.